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BUSBAR MODULE INCLUDING CLAMPING AND GUIDING FEATURES

Abstract

A battery pack may include a busbar including a slot. The battery pack may further include a battery cell including a housing and a tab terminal extending from the housing. The tab terminal may be received within the slot such that at least a portion of the tab terminal is positioned on an opposite side of the busbar from the housing. The battery pack may additionally include a busbar frame including a projection. The projection may be received within the slot such that at least a portion of the projection is positioned on the opposite side of the busbar from the housing.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to electrified vehicles, and more specifically relates to a busbar module including clamping and guiding features.

BACKGROUND

[0002] A high voltage traction battery pack typically powers the electric machines and other electrical loads of an electrified vehicle. The traction battery pack includes a plurality of battery cells. The battery cells must be reliably connected to one another in order to provide the voltage and power levels necessary for achieving vehicle propulsion.

SUMMARY

[0003] In some aspects, the techniques described herein relate to a battery pack, including: a busbar including a slot; a battery cell including a housing and a tab terminal extending from the housing, wherein the tab terminal is received within the slot such that at least a portion of the tab terminal is positioned on an opposite side of the busbar from the housing; and a busbar frame including a projection, wherein the projection is received within the slot such that at least a portion of the projection is positioned on the opposite side of the busbar from the housing.

[0004] In some aspects, the techniques described herein relate to a battery pack, wherein the tab terminal is in direct contact with a surface of a bent portion of the busbar.

[0005] In some aspects, the techniques described herein relate to a battery pack, wherein the bent portion establishes a base of the slot.

[0006] In some aspects, the techniques described herein relate to a battery pack, wherein the tab terminal is also in direct contact with the projection.

[0007] In some aspects, the techniques described herein relate to a battery pack, wherein a surface of the projection in direct contact with the tab terminal includes at least one rib.

[0008] In some aspects, the techniques described herein relate to a battery pack, wherein a surface of the projection opposite the surface in direct contact with the tab terminal includes a rounded profile.

[0009] In some aspects, the techniques described herein relate to a battery pack, wherein an end of the tab terminal projects further into the slot than the projection, such that the end of the tab terminal is spaced-apart further from the side of the busbar opposite the housing than an end of the projection.

[0010] In some aspects, the techniques described herein relate to a battery pack, wherein: the busbar frame includes a main section, the main section includes a slot, and the tab terminal projects through the slot of the busbar frame and through the slot of the busbar.

[0011] In some aspects, the techniques described herein relate to a battery pack, wherein the busbar frame further includes a first ramped surface on a first side of the slot and projecting toward the housing, and a second ramped surface on a second side of the slot and projecting toward the housing.

[0012] In some aspects, the techniques described herein relate to a battery pack, wherein the first and second ramped surfaces face toward one another.

[0013] In some aspects, the techniques described herein relate to a battery pack, wherein: the first and second ramped surfaces are provided by first and second projections, respectively, projecting toward the housing.

[0014] In some aspects, the techniques described herein relate to a battery pack, wherein the projection includes at least one finger projecting further from the side of the busbar opposite the housing than a remainder of the projection.

[0015] In some aspects, the techniques described herein relate to a battery pack, wherein the at least one finger includes a first side finger arranged adjacent a first side edge of the projection, and a second side finger arranged adjacent a second side edge of the projection opposite the first side.

[0016] In some aspects, the techniques described herein relate to a battery pack, wherein the at least

one finger further includes a third finger arranged substantially between the first and second side fingers.
[0017] In some aspects, the techniques described herein relate to a battery pack, including a weld that secures the tab terminal to the bent portion.
[0018] In some aspects, the techniques described herein relate to a battery pack, wherein the busbar frame and the busbar establish a busbar module of the battery pack.
[0019] In some aspects, the techniques described herein relate to a method, including: electrically connecting a tab terminal of a battery cell to a busbar by inserting the tab terminal through a busbar frame and through a slot in the busbar, wherein the busbar frame includes a projection, wherein the projection is received within the slot such that at least a portion of the projection is positioned on an opposite side of the busbar from a housing of the battery cell.
[0020] In some aspects, the techniques described herein relate to a method, wherein the tab terminal is in direct contact with the projection and a bent portion of the busbar.
[0021] In some aspects, the techniques described herein relate to a method, wherein: the busbar frame includes a main section, the main section includes a slot, and the tab terminal projects through the slot of the busbar frame and through the slot of the busbar.
[0022] In some aspects, the techniques described herein relate to a method, wherein: the busbar frame further includes a first ramped surface on a first side of the slot and projecting toward the housing, and a second ramped surface on a second side of the slot and projecting toward the housing, and the first and second ramped surfaces guide the tab terminal as the tab terminal is inserted through the busbar frame.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0023] FIG. **1** schematically illustrates an electrified vehicle.
[0024] FIG. **2** is a perspective view of a battery pack for an electrified vehicle.
[0025] FIG. **3** is a cross-sectional view of the battery pack of FIG. **2**.
[0026] FIG. **4** illustrates a battery cell of a battery array of a battery pack.
[0027] FIG. **5** illustrates an example busbar module relative to a battery array.
[0028] FIG. **6** is a cross-sectional view through section **6-6** of FIG. **5**.
[0029] FIG. **7** is a close-up, perspective view of the busbar module, in which a projection of a busbar frame has two fingers.
[0030] FIG. **8** is a close-up, perspective view of the busbar module, in which the projection of the busbar frame has three fingers.
[0031] FIG. **9** is a close-up, perspective view of the busbar module, in which the projection of the busbar frame is without fingers.
[0032] FIG. **10** is a cross-sectional view similar to FIG. **6** and illustrates an example configuration of the projection including a rib.
DETAILED DESCRIPTION
[0033] This disclosure relates generally to electrified vehicles, and more specifically relates to a busbar module including clamping and guiding features. A corresponding method is also disclosed. The busbar module of this disclosure facilitates assembly of a battery pack by permitting one to readily align multiple tab terminals relative to a busbar. The busbar module of this disclosure further facilitates assembly of the battery pack by holding the tab terminals in place relative to a corresponding portion of the busbar. These and other benefits will be appreciated from the following description.
[0034] FIG. **1** schematically illustrates an electrified vehicle **10**. The electrified vehicle **10** may include any type of electrified powertrain. In an embodiment, the electrified vehicle **10** is a battery

electric vehicle (BEV). However, the concepts described herein are not limited to BEVs and could extend to other electrified vehicles, including, but not limited to, hybrid electric vehicles (HEVs), plug-in hybrid electric vehicles (PHEV's), fuel cell vehicles, etc. Therefore, although not specifically shown in the exemplary embodiment, the powertrain of the electrified vehicle **10** could be equipped with an internal combustion engine that can be employed either alone or in combination with other power sources to propel the electrified vehicle **10**.

[0035] In the illustrated embodiment, the electrified vehicle **10** is depicted as a car. However, the electrified vehicle **10** could alternatively be a sport utility vehicle (SUV), a van, a pickup truck, or any other vehicle configuration.

[0036] In the illustrated embodiment, the electrified vehicle **10** is a full electric vehicle propelled solely through electric power, such as by one or more electric machines **12**, without assistance from an internal combustion engine. The electric machine **12** may operate as an electric motor, an electric generator, or both. The electric machine **12** receives electrical power and can convert the electrical power to torque for driving one or more wheels **14** of the electrified vehicle **10**.

[0037] A voltage bus **16** may electrically couple the electric machine **12** to a traction battery pack **18**. The traction battery pack **18** is an exemplary electrified vehicle battery. The traction battery pack **18** may be a high voltage traction battery pack assembly that includes a plurality of battery cells capable of outputting electrical power to power the electric machine **12** and/or other electrical loads of the electrified vehicle **10**. Other types of energy storage devices and/or output devices could alternatively or additionally be used to electrically power the electrified vehicle **10**.

[0038] The traction battery pack **18** may be secured to an underbody **20** of the electrified vehicle **10**. However, the traction battery pack **18** could be located elsewhere on the electrified vehicle **10** within the scope of this disclosure.

[0039] FIGS. **2** and **3** illustrate additional details associated with the traction battery pack **18** of the electrified vehicle **10**. The traction battery pack **18** may include one or more battery arrays **22** (e.g., battery assemblies or groupings of rechargeable battery cells **24**) capable of outputting electrical power to power the electric machine **12** and/or other electrical loads of the electrified vehicle **10**. Other types of energy storage devices and/or output devices could alternatively or additionally be used to electrically power the electrified vehicle **10**.

[0040] The battery cells **24** may be stacked side-by-side along a stack axis to construct a grouping of battery cells **24**, sometimes referred to as a “cell stack.” In the highly schematic depiction of FIG. **3**, the battery cells **24** are stacked in a direction into the page to construct each battery array **22**, and thus the battery arrays **22** may extend in cross-car direction. However, other configurations may also be possible. The total number of battery arrays **22** and battery cells **24** provided within the traction battery pack **18** is not intended to limit this disclosure.

[0041] In an embodiment, the battery cells **24** of each battery array **22** are pouch style, lithium-ion cells. However, battery cells having other geometries (cylindrical, prismatic, etc.), other chemistries (nickel-metal hydride, lead-acid, etc.), or both could alternatively be utilized within the scope of this disclosure.

[0042] The battery arrays **22** and various other battery internal components (e.g., bussed electrical center, battery electric control module, wiring, connectors, etc.) may be housed within an interior area **26** of an enclosure assembly **28**. The enclosure assembly **28** may include an enclosure cover **30** and an enclosure tray **32**. The enclosure cover **30** may be secured (e.g., bolted, welded, adhered, etc.) to the enclosure tray **32** to provide the interior area **26**. The size, shape, and overall configuration of the enclosure assembly **28** is not intended to limit this disclosure.

[0043] FIG. **4** illustrates one of the battery cells **24** that can be provided within the traction battery pack **18**. Each battery cell **24** may include a housing **36** and a pair of tab terminals **34** that project outwardly from the housing **36**. In an embodiment, each battery cell **24** includes two tab terminals **34**, with one tab terminal **34** protruding outwardly at each opposing side of the housing **36**. In some implementations, the housing **36** is a multi-layered structure including an aluminum film layer and a polymer layer. However, other housing configurations are also contemplated within the scope of this disclosure.

[0044] The tab terminals **34** of the battery cells **24** of each battery array **22** must be reliably connected to one another in order to provide the voltage and power levels necessary for achieving vehicle propulsion. Busbars are sometimes used for making such a connection, however, it can be difficult to properly position and align the tab terminals **34** relative to the busbar during assembly and welding processes. This disclosure is therefore specifically directed to busbars that include features for facilitating proper positioning and alignment of the tab terminals **34** when joining the tab terminals **34** to a busbar.

[0045] FIGS. **5** and **6** illustrate an example busbar module **37** according to this disclosure relative to a grouping of battery cells **24**. The example busbar module **37** includes a busbar **38** and a busbar frame **39**.

[0046] The busbar **38** may be joined to tab terminals **34** of a grouping of battery cells **24** for electrically connecting the battery cells **24**. The grouping of battery cells **24** may be part of one of the battery arrays **22** of the traction battery pack **18**, for example. Once electrically coupled, the battery cells **24** may supply electrical power for powering various components of the electrified vehicle **10**. In the illustrated embodiment, the busbar **38** is configured to join the tab terminals **34** of seven adjacent battery cells **24**. However, the busbar **38** could be configured to join a greater or fewer number of tab terminals **34**/battery cells **24** within the scope of this disclosure.

[0047] The busbar **38** may include a plurality of slots **40**. Each slot **40** may be formed through a main body **45** of the busbar **38** and may be sized for receiving one of the cell tab terminals **34**. In an embodiment, each slot **40** includes a first width W**1** that is larger than a second width W**2** of the cell tab terminal **34** received therethrough. The cell tab terminals **34** may extend through the slots **40** such that at least a portion of each cell tab terminal **34** is located on an opposite side of the busbar **38** from the housing **36** of its respective battery cell **24**.

[0048] The busbar **38** may additionally include a plurality of bent portions **42**. One bent portion **42** may be provided adjacent each slot **40** of the busbar **38**, and the bent portion **42** may establish a base of the slot **40**, for example. The bent portions **42** are bent in a direction away from the battery cells **24**, namely away from the housings **36** of the battery cells **24**.

[0049] Each slot **40** may be formed in a punch or cutting process, and each bent portion **42** may be formed in a separate bending process. Alternatively, each slot **40** and bent portion **42** pair may be formed in a single step as part of a lance and form process.

[0050] The tab terminals **34** may be received against and supported by an upper surface **44** of the bent portions **42**. Each upper surface **44** may provide a relatively flat surface for locating the tab terminal **34** relative to the respective slot **40** and bent portion **42**.

[0051] The tab terminals **34** may be joined to the upper surface **44** of the bent portion **42** by one or more welds **46**. The welds **46** may be linear or non-linear welds. Prior to formation of the welds **46**, the tab terminals **34** may be clamped and/or spot welded to the bent portions **42** to aid in the joining process.

[0052] The busbar frame **39** is made of an electrically insulative material, in this example. The busbar frame **39** includes features configured to facilitate clamping of a tab terminal **34** relative to a respective bent portion **42**, and to guide the tab terminals **34** relative to a respective slot **40**.

[0053] One clamping assembly will now be described. With reference to FIG. **6**, busbar frame **39** includes a main body **48** arranged on the same side of the busbar **38** as the housings **36** of the battery cells **24**. The busbar frame **39** further includes a projection **50** extending through the slot **40** such that a portion, namely an end **52**, of the projection **50** is on a side of the busbar opposite the housings **36**.

[0054] The projection **50** is arranged such that the upper surface of the tab terminal **34** is in direct contact with a bottom surface **54** of the projection **50**. The bottom surface **54** is substantially parallel to upper surface **44**. In this way, the projection **50** is configured to clamp the tab terminal

34 to the bent portion **42**. In this disclosure, the terms “upper” and “bottom” are used with reference to the orientation of the components in FIG. **6**, for example, and are not otherwise limiting.

[0055] The bottom surface **54** of the projection **50** may include one or more raised surface features to facilitate engagement with the tab terminal **34**, such as one or more ribs or ridges. An example rib **55** is shown in FIG. **10** projecting downward from the remainder of the bottom surface **54** of projection **50**. In this example, the rib **55** is provided in the location of a first side finger **60** (explained below). The rib **55** could be in another location along the projection **50** and is not required to be on a finger. An upper surface **56** of the projection **50** adjacent the end **52** exhibits a rounded profile in this example to facilitate positioning the projection **50** into the slot **40**.

[0056] In this disclosure, the end **58** of the tab terminal **34** is spaced-apart further from the main body **45** of the busbar **38** than the end **52**. In this way, the projection **50** does not interfere with the formation of welds **46**.

[0057] The projection **50** may include one or more fingers projecting further through the slot **40** than other portions of the projection **50**, as shown in FIGS. **7** and **8**. In FIG. **7**, the projection **50** includes a first side finger **60** arranged adjacent a first side edge **62** of the projection **50**, and a second side finger **64** arranged adjacent a second side edge **66** of the projection **50** opposite the first side. In FIG. **8**, the projection includes a third finger **68** arranged substantially between the first and second side fingers **60**, **64**. When the projection **50** includes one or more fingers, the welds **46** may be provided adjacent, or between, those fingers. The fingers may be formed, in one example, by selectively cutting certain portions of the projection **50**.

[0058] FIG. **9** illustrates an embodiment in which the projection **50** does not exhibit any fingers, and instead exhibits a constant profile along its length. While some example projections have been described, it should be understood that the busbar frame **39** includes a plurality of such projections, with one projection corresponding to each slot **40**.

[0059] The busbar frame **39** further includes a plurality of slots **70** formed in the main body **48**. The slots **70** are aligned with a corresponding slot **40** of the busbar **38**. The busbar frame **39** includes a plurality of guiding features configured to facilitate guiding of the tab terminals relative to the slots **40**, **70**, with one arrangement of guiding features provided adjacent each of the slots **70**. An example guiding feature will now be described.

[0060] The busbar frame **39** includes a first guide projection **72** and a second guide projection **74** projecting from main body **48** and toward the housing **36**. The first and second guide projections **72**, **74** are arranged adjacent the slot **70** and are provided on opposite sides of the slot **70**. The first guide projection **72** includes a first ramped surface **76** facing toward the slot **70** and moving gradually away from the slot **70** as the first ramped surface **76** moves toward the housing **36**. The first guide projection **74** includes a second ramped surface **78** facing toward the slot **70** and toward the first ramped surface **76**. The second ramped surface **78** moves gradually away from the slot **70** as the second ramped surface **78** moves toward the housing **36**. If the tab terminal **34** comes into contact with either of the first or second ramped surfaces **76**, **78** during insertion of the tab terminal **34** into the slot **70**, the first and second ramped surfaces **76**, **78** naturally guide the tab terminal **34** back toward the slot **70**.

[0061] The busbar **38** may be a metallic component of the busbar module **37**, and the busbar frame **39** may be a plastic component of the busbar module **37**. In an embodiment, the busbar **38** is made of copper or aluminum, and the busbar frame **39** is made of polypropylene or polyethylene. However, other materials are contemplated within the scope of this disclosure.

[0062] The embodiments described above and shown in FIGS. **5** and **6** provide a “lap joint” design between the busbar **38** and the tab terminals **34**. However, other configurations are contemplated within the scope of this disclosure.

[0063] It should be understood that terms such as “about” and “substantially” are not intended to be boundaryless terms, and should be interpreted consistent with the way one skilled in the art would

interpret those terms. Directional terms such as “above,” “upper,” “below,” “bottom,” etc., are used with reference to the arrangement of the corresponding components in the drawings and are not intended to otherwise be limiting.
[0064] Although the different examples have the specific components shown in the illustrations, embodiments of this disclosure are not limited to those particular combinations. It is possible to use some of the components or features from one of the examples in combination with features or components from another one of the examples.
[0065] One of ordinary skill in this art would understand that the above-described embodiments are exemplary and non-limiting. That is, modifications of this disclosure would come within the scope of the claims. Accordingly, the following claims should be studied to determine their true scope and content.

Claims

1. A battery pack, comprising: a busbar including a slot; a battery cell including a housing and a tab terminal extending from the housing, wherein the tab terminal is received within the slot such that at least a portion of the tab terminal is positioned on an opposite side of the busbar from the housing; and a busbar frame including a projection, wherein the projection is received within the slot such that at least a portion of the projection is positioned on the opposite side of the busbar from the housing.
2. The battery pack as recited in claim 1, wherein the tab terminal is in direct contact with a surface of a bent portion of the busbar.
3. The battery pack as recited in claim 2, wherein the bent portion establishes a base of the slot.
4. The battery pack as recited in claim 2, wherein the tab terminal is also in direct contact with the projection.
5. The battery pack as recited in claim 4, wherein a surface of the projection in direct contact with the tab terminal includes at least one rib.
6. The battery pack as recited in claim 4, wherein a surface of the projection opposite the surface in direct contact with the tab terminal includes a rounded profile.
7. The battery pack as recited in claim 2, wherein an end of the tab terminal projects further into the slot than the projection, such that the end of the tab terminal is spaced-apart further from the side of the busbar opposite the housing than an end of the projection.
8. The battery pack as recited in claim 1, wherein: the busbar frame includes a main section, the main section includes a slot, and the tab terminal projects through the slot of the busbar frame and through the slot of the busbar.
9. The battery pack as recited in claim 8, wherein the busbar frame further includes a first ramped surface on a first side of the slot and projecting toward the housing, and a second ramped surface on a second side of the slot and projecting toward the housing.
10. The battery pack as recited in claim 9, wherein the first and second ramped surfaces face toward one another.
11. The battery pack as recited in claim 9, wherein: the first and second ramped surfaces are provided by first and second projections, respectively, projecting toward the housing.
12. The battery pack as recited in claim 1, wherein the projection includes at least one finger projecting further from the side of the busbar opposite the housing than a remainder of the projection.
13. The battery pack as recited in claim 12, wherein the at least one finger includes a first side finger arranged adjacent a first side edge of the projection, and a second side finger arranged adjacent a second side edge of the projection opposite the first side.
14. The battery pack as recited in claim 13, wherein the at least one finger further includes a third finger arranged substantially between the first and second side fingers.

15. The battery pack as recited in claim 2, comprising a weld that secures the tab terminal to the bent portion.
16. The battery pack as recited in claim 1, wherein the busbar frame and the busbar establish a busbar module of the battery pack.
17. A method, comprising: electrically connecting a tab terminal of a battery cell to a busbar by inserting the tab terminal through a busbar frame and through a slot in the busbar, wherein the busbar frame includes a projection, wherein the projection is received within the slot such that at least a portion of the projection is positioned on an opposite side of the busbar from a housing of the battery cell.
18. The method as recited in claim 17, wherein the tab terminal is in direct contact with the projection and a bent portion of the busbar.
19. The method as recited in claim 17, wherein: the busbar frame includes a main section, the main section includes a slot, and the tab terminal projects through the slot of the busbar frame and through the slot of the busbar.
20. The method as recited in claim 19, wherein: the busbar frame further includes a first ramped surface on a first side of the slot and projecting toward the housing, and a second ramped surface on a second side of the slot and projecting toward the housing, and the first and second ramped surfaces guide the tab terminal as the tab terminal is inserted through the busbar frame.